

Engineered basement membrane matrices to study morphogenesis and invasion of mammary epithelium

Jane Baude¹, Daniella Walter², Marissa Gionet-Gonzales², Ryan Stowers^{2,3}

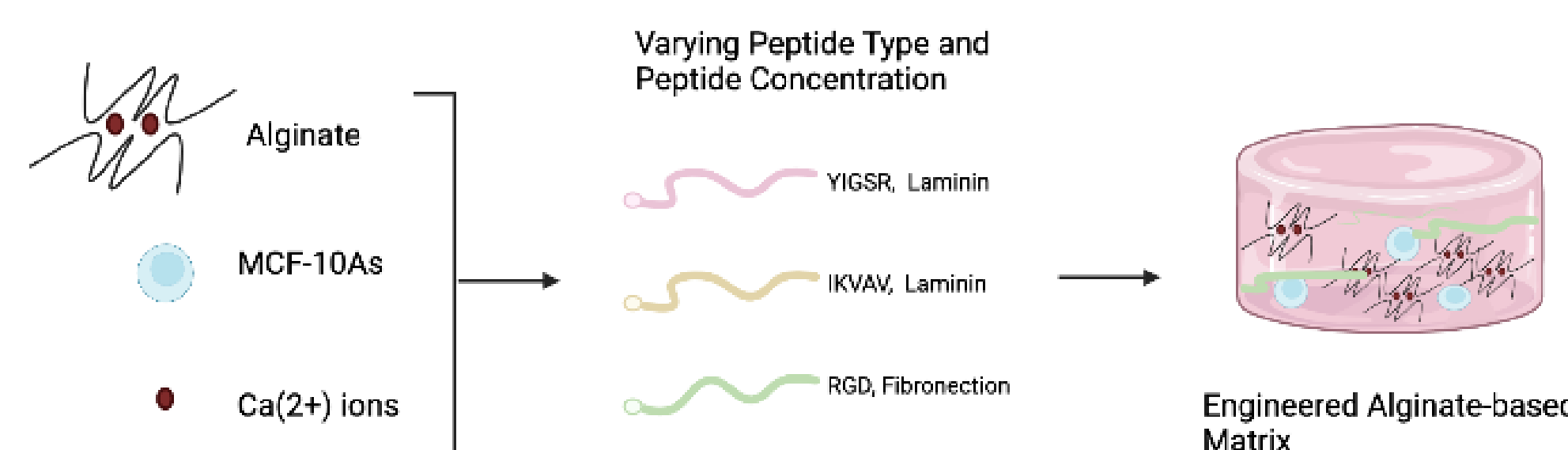
¹ Dept. of Molecular, Cellular, and Developmental Biology, UCSB, ² Dept. of Mechanical Engineering, ³ Dept. of Biological Engineering

Introduction

Matrigel, reconstituted basement membrane composed primarily of laminin, entactin, fibronectin, and collagen IV, is used as an ECM surrogate for a vast variety of cell culture applications¹. Despite its widespread use, Matrigel has intrinsic limitations¹, including:

- Lack of tunability
- Inconsistencies between batches
- Xenogenic contaminants
- Chronic supply shortages

We hypothesize that by conjugating ECM peptides to an alginate matrix, we will be able to create a tunable engineered basement membrane. This system would not only serve as a replacement for Matrigel but would also allow for superior tunability. To test this hypothesis, we will evaluate whether this system can support mammary acinar morphogenesis of MCF-10As, a nonmalignant mammary epithelium cell line. We will also test the impact of stiffness and ligand concentration using this system.



Created in BioRender.com²

Figure 1: Using carbodiimide chemistry, ECM peptide motifs (IKVAV, YIGSR, and RGD) were conjugated to alginate at varying concentrations^{3,4} in order to create engineered reductionist matrices. These ligands were chosen based off availability. Matrix stiffnesses “soft”, “medium” and “stiff” were chosen to represent these *in vivo* tissue elastic moduli⁵. MCF-10As were encapsulated, cultured for 14 days, and then analyzed to compare acinus formation versus invasion. Created with BioRender.com².

Mammary Acinus Formation

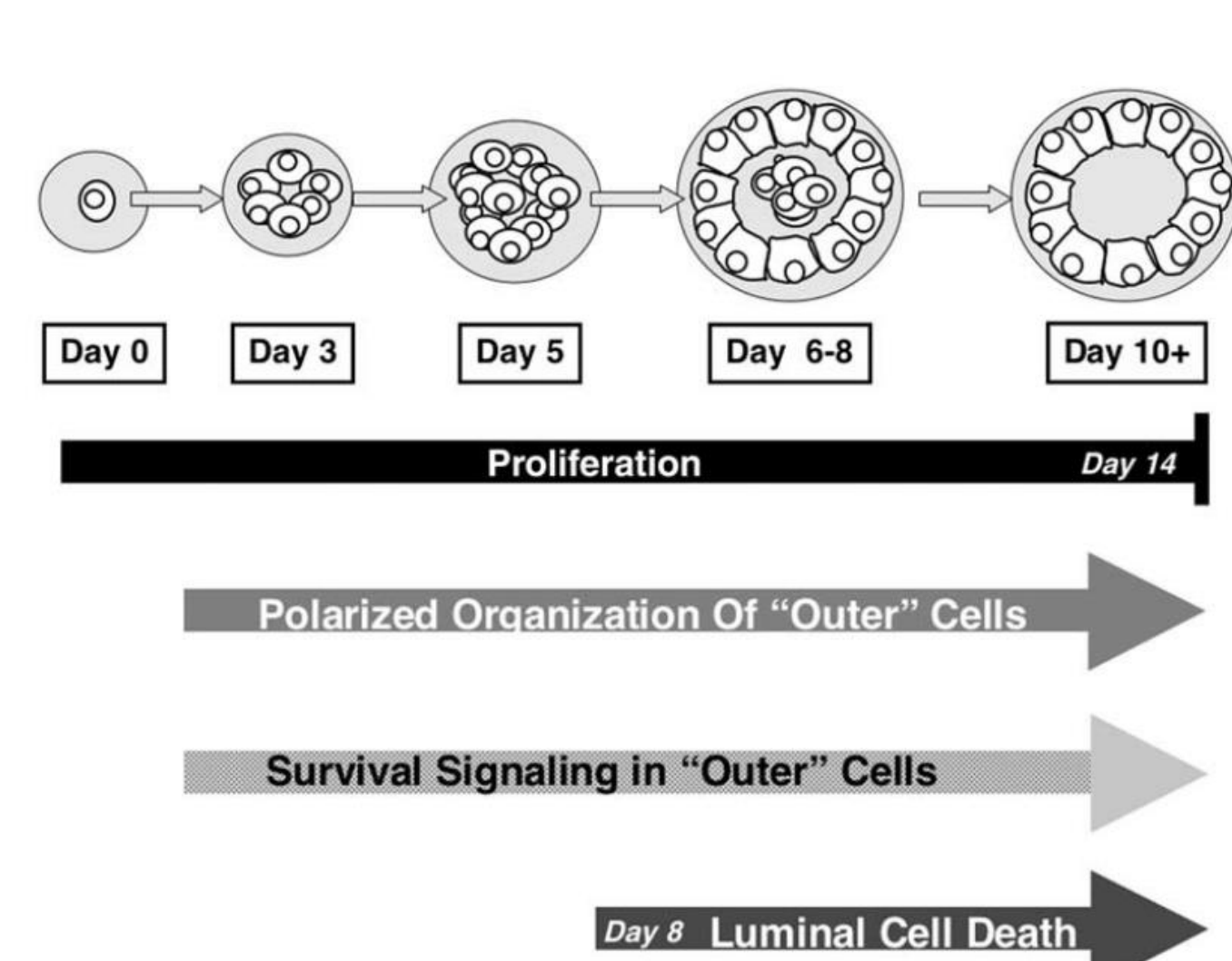


Figure 2: MCF-10A acinar morphogenesis from a single cell to a glandular, polarized acinus⁴. Mammary epithelium cells develop into their healthy, acini structure when cultured within 3D soft (~100 Pa) culture systems⁵, but the cells become invasive and tumorigenic in 3D stiff (>2,000 Pa) culture systems. (Figure reproduced from: Debnath, J & Brugge, J.S, *Nature Rev. Cancer*, (2005))

Acinus Formation in Engineered Matrices

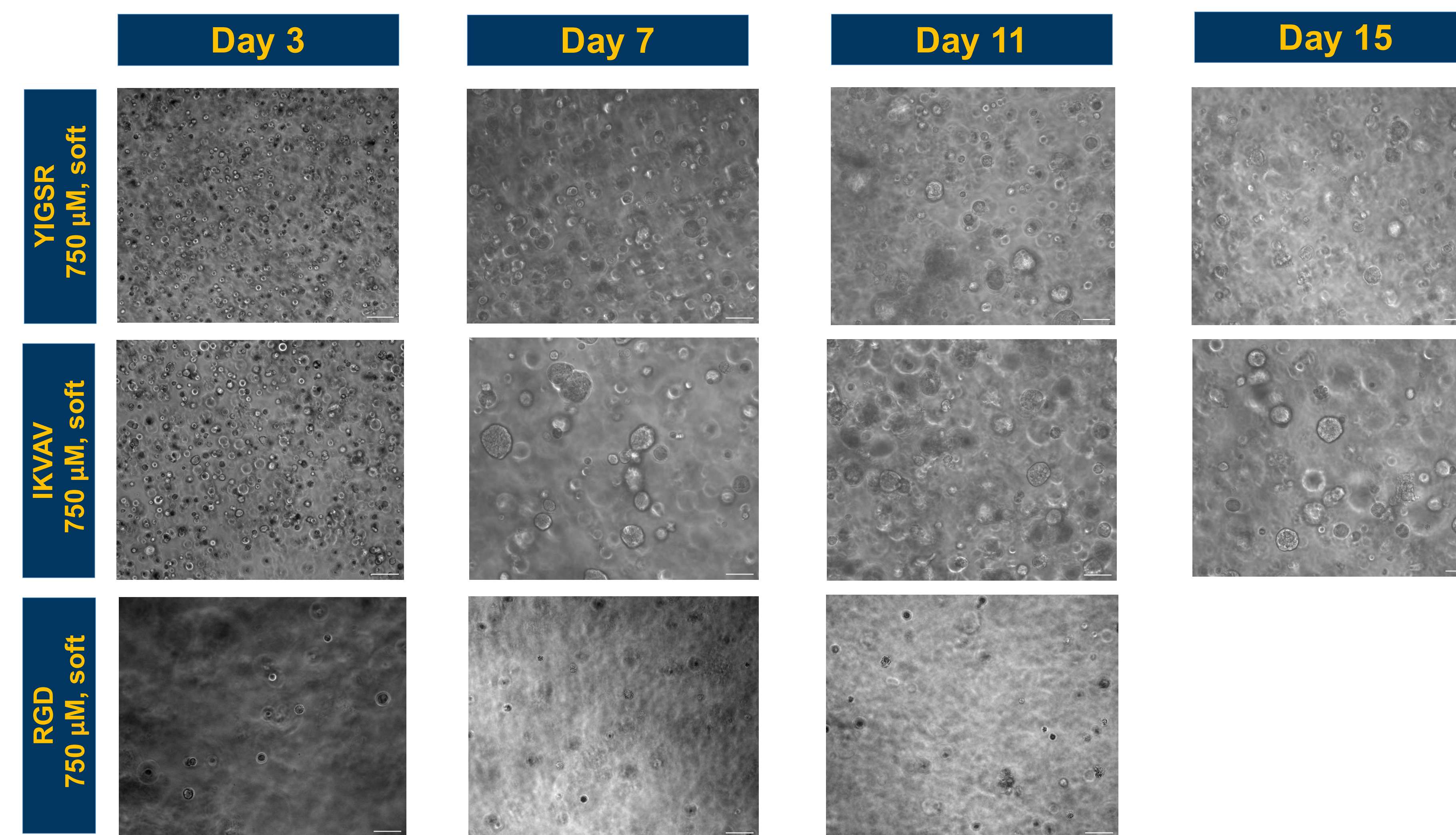
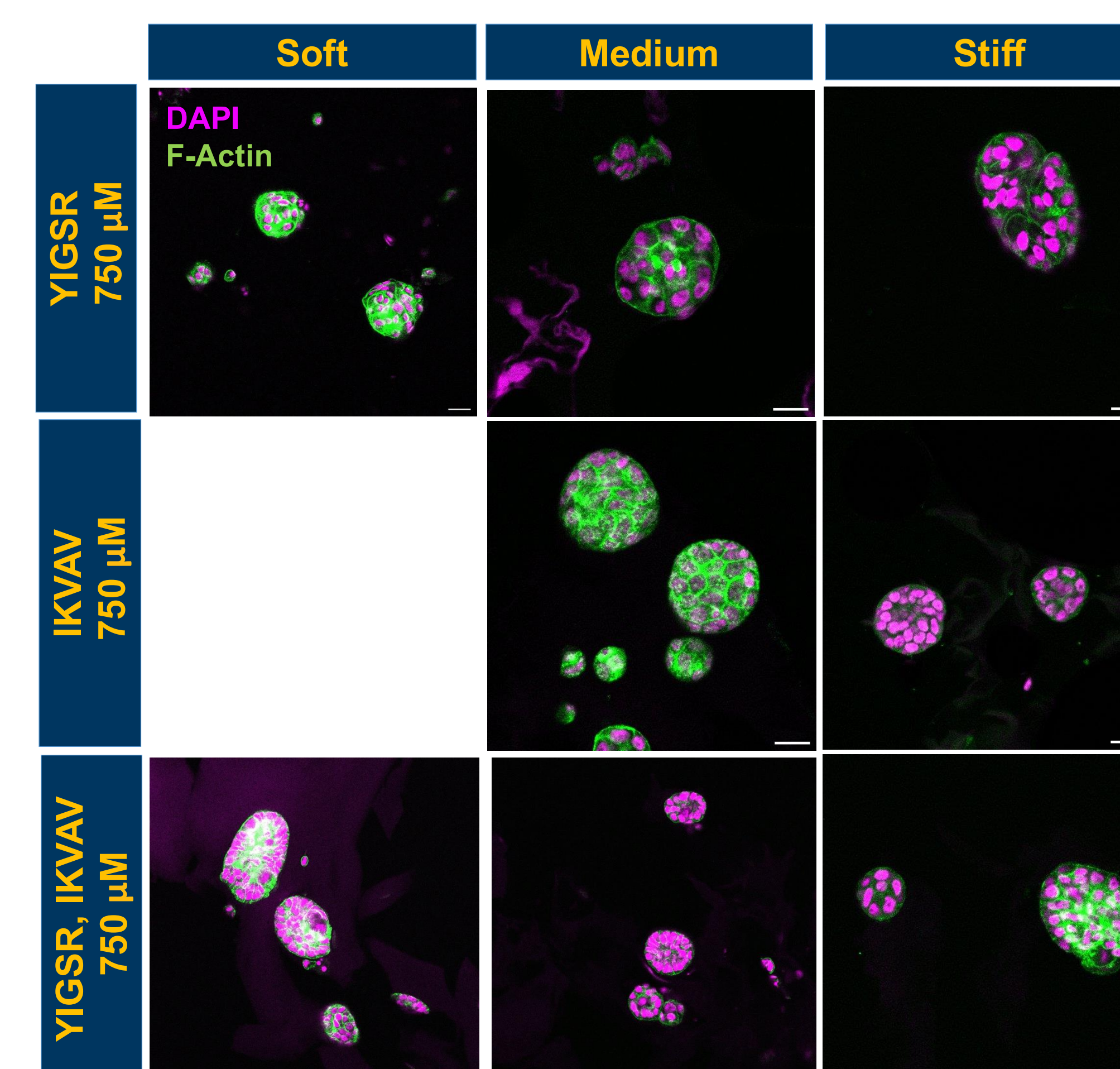
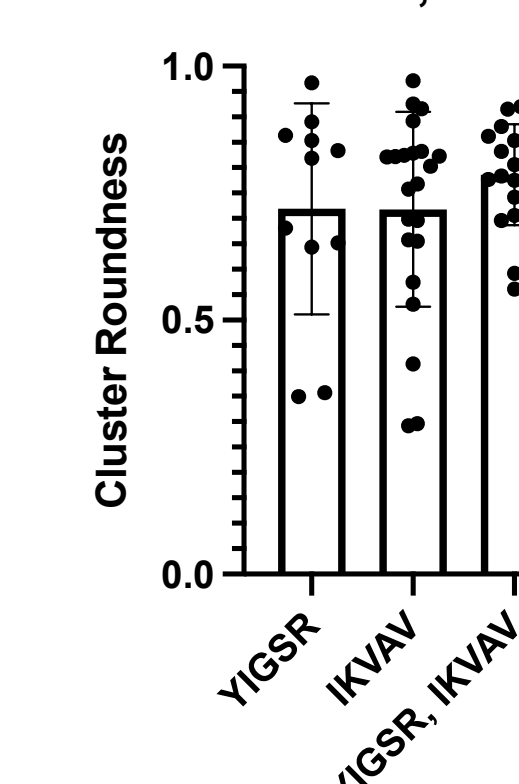


Figure 3: Brightfield images showing growth of MCF-10As encapsulated in their respective gel condition. In soft YIGSR and IKVAV matrices, single cells develop into acini by the end of the two-week culture period. Cells do not cluster in RGD matrices. Scale bars = 10000 μ M. Photos taken at 10X.

Elevated Stiffness Alone does not Induce Invasion



YIGSR versus YIGSR,IKVAV (stiff, 750 μ M)



YIGSR versus YIGSR,IKVAV (soft, 750 μ M)

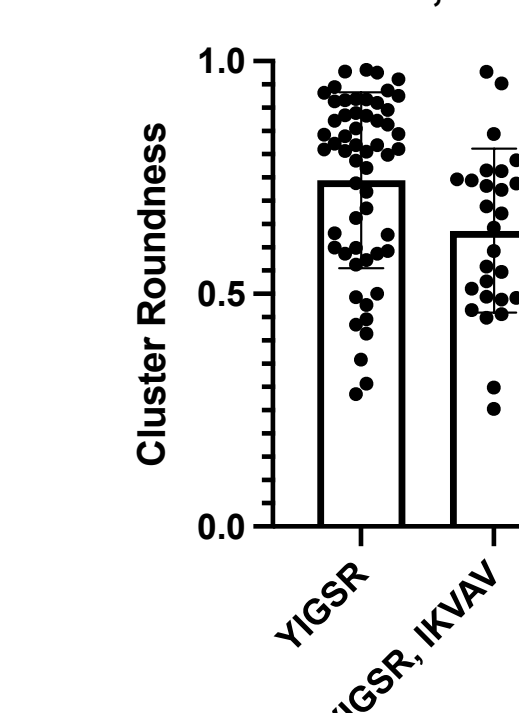


Figure 4: Left: MCF-10As in fast relaxing 750 μ M YIGSR stiff, medium, and soft matrices, respectively. Generally, we were able to obtain acinar looking cell clusters, regardless of stiffness of the matrix. Scale bar = 25 μ M. Right: Analysis of cell roundness from collective confocal images.

Matrix Stiffness and Ligand Concentration Combine to Regulate Invasion

Here we have begun to investigate how stiffness and ligand concentration can give rise to invasion. In our engineered basement membrane, MCF10As displayed different morphologies depending on matrix stiffness, and ligand concentration. Of importance, it was seen that high stiffness alone is not enough to induce invasion in our 3D system. We were only able to obtain the malignant phenotype when there was a stiff matrix + low concentration of the ligand.

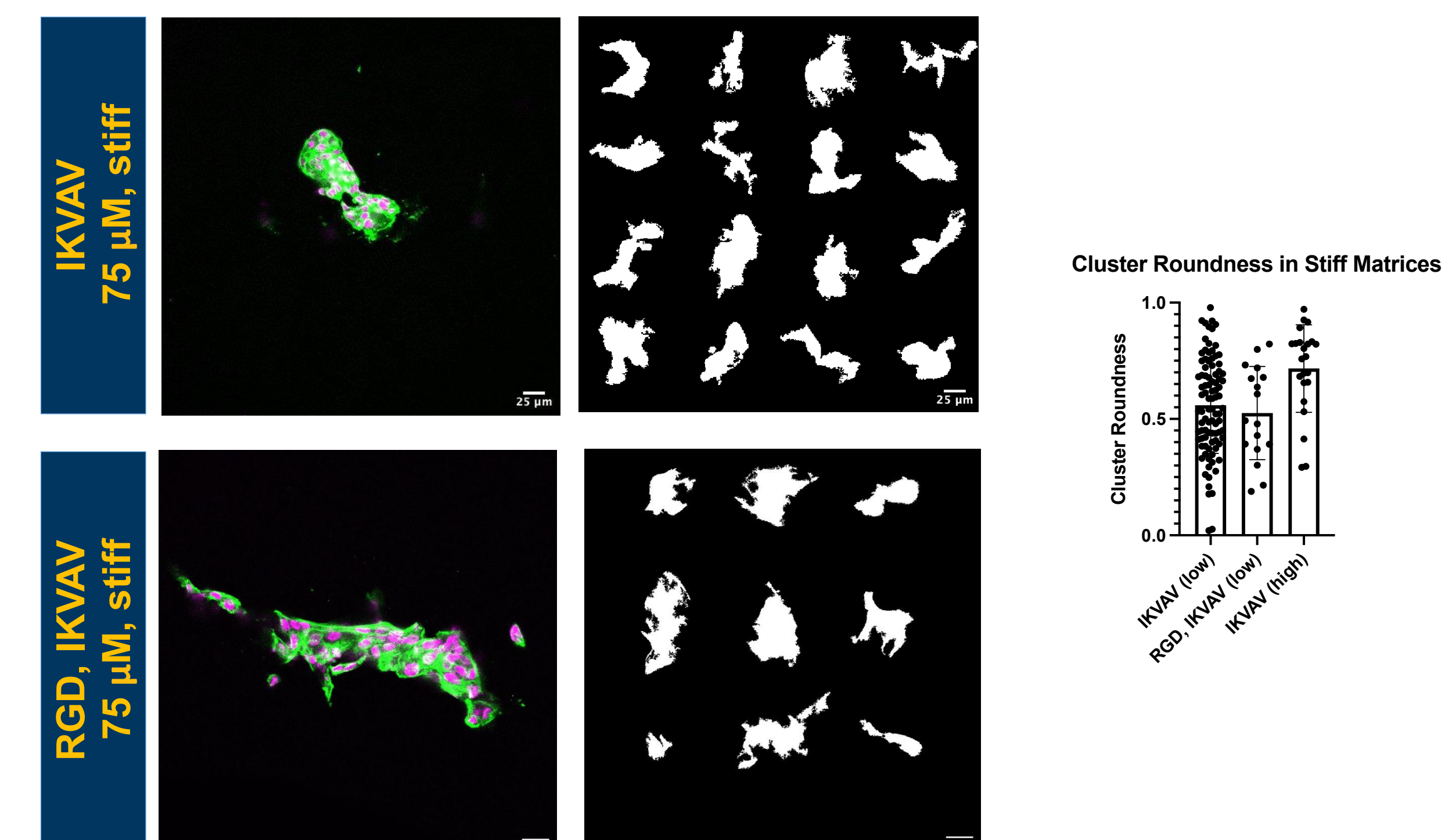


Figure 5: Left: Confocal Images showing invasive clusters in stiff gel matrices with a low (75 μ M) peptide concentration. Scale bars = 25 μ M. Right: Analysis of cell roundness from collective confocal images comparing stiff IKVAV matrices with low and high ligand concentration.

Conclusions

- This alginate based system aims to not only replace Matrigel, but to also provide mechanical and biochemical tunability.
- These results demonstrate a proof-of-concept that reductionist matrices are sufficient to induce 3D mammary acinar morphogenesis of MCF-10A cells.
- High matrix stiffness (>2,000 Pa) alone was not enough to drive invasion of MCF-10As.
- Invasion of MCF-10As was only seen when there was both a low ligand concentration and a stiff matrix.
- This is an excellent platform to decipher how cell-matrix interactions lead to phenotypic outcomes.

Acknowledgements + References

1. Aisenbrey, E. A., & Murphy, W. L. *Nature Reviews Materials*, (2020).
2. Created with BioRender.com
3. Chaudhuri, O., K. *Nature materials*, (2014).
4. Sawicki, L. A., et al., *APL Bioengineering*, (2019).
5. Debnath, J. & Brugge, J. S. *Nature Rev. Cancer*, (2005).

UC Cancer Research Coordinating Committee C22CR4078
T32 Quantitative Mechanobiology Training Grant, NIH, 1T32GM141846